

Predictive Model Plan

Geldium Finance: Credit Card Delinquency Risk | Prepared by Tata iQ. Model logic was drafted with GenAI support, then reviewed and adapted using the Task 1 EDA findings.

1. Model Logic (Generated with GenAI)

Objective: predict which Geldium customers are likely to become delinquent (Delinquent_Account = 1) so the Collections team can prioritise early, targeted outreach.

Chosen model (summary):

I would use a regularised logistic regression as the primary model, with a shallow decision tree as a transparent challenger and a gradient-boosted model as an optional benchmark. Logistic regression turns customer attributes into a delinquency probability that can be explained feature by feature, which suits a regulated lending environment.

Top 5 input features:

- **Payment history (Month_1 to Month_6, Missed_Payments):** engineered into counts of late and missed months and a recent-months flag; the most direct behavioural signal of repayment stress.
- **Credit_Utilization:** high use of available credit signals financial pressure.
- **Debt_to_Income_Ratio:** measures capacity to service existing debt.
- **Credit_Score:** summary of past credit behaviour.
- **Employment_Status (with Income and Account_Tenure as supporting features):** captures income stability and how established the relationship is.

These are chosen on credit-risk logic. The Task 1 EDA found weak linear links to delinquency in this dataset, so the model will confirm empirically which features actually carry signal.

How the model processes data (workflow):

Stage	What happens
1. Data ingestion	Load customer records; drop Customer_ID from the feature set and keep it only as a key.
2. Cleaning (from EDA)	Standardise Employment_Status labels; convert utilization and DTI to percentages; encode Month_1-6 as 0/1/2; median-impute Income, Loan_Balance and Credit_Score with missing flags (fitted on training data

	only).
3. Feature engineering	Late_Months, Missed_Months, Recent_Miss (last 2 months), missing-value flags; scale numeric variables; one-hot encode categories.
4. Split	Stratified train/validation/test split (or 5-fold cross-validation) because only 16% of customers are delinquent.
5. Training	Fit L2-regularised logistic regression with class weights to handle imbalance; tune regularisation on validation data.
6. Prediction output	Probability of delinquency per customer, mapped to risk tiers (Low / Medium / High) with the top contributing factors.
7. Action layer	High-risk tiers routed to Collections for early intervention (reminders, payment plans); a human reviews high-impact decisions.

Pseudocode (GenAI-assisted, paraphrased):

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load data -> df
clean: standardise labels, rescale utilization/DTI, encode Month_1..Month_6
engineer: Late_Months, Missed_Months, Recent_Miss, missing flags
X = features (exclude Customer_ID, Location); y = Delinquent_Account
split X, y into train / test (stratified on y)
pipeline = impute(median) -> scale -> one-hot ->
LogisticRegression(class_weight='balanced')
tune regularisation strength with cross-validation on train
fit pipeline on train
p = pipeline.predict_proba(test)[: , 1]      # delinquency probability
tier = Low if p < t1; Medium if t1 <= p < t2; High if p >= t2
evaluate: AUC, precision, recall, F1, calibration, fairness by group
explain: coefficients / odds ratios (and SHAP for any complex model)

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2. Justification for Model Choice

I recommend logistic regression because it best balances predictive power, transparency and operational simplicity for Geldium's Collections team. Its coefficients show how each factor (for example utilization or missed payments) raises or lowers delinquency risk, which makes decisions easy to explain to the Head of Collections, to customers and to regulators, and supports adverse-action and fair-lending reviews. It trains quickly, handles Geldium's mix of numerical and encoded categorical data, is simple to deploy and monitor, and gives calibrated probabilities that can drive risk tiers and prioritise limited outreach capacity. A neural network or boosted trees may capture non-linear interactions and can be benchmarked, but they are harder to explain and audit and offer little advantage on a 500-record dataset.

Model comparison:

Model	Strengths	Trade-offs	Fit for Geldium
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Logistic regression (chosen)	Highly interpretable; fast; calibrated probabilities; standard in credit risk; easy to monitor.	Assumes mostly linear effects; needs manual interaction terms.	Best: transparent and regulator-friendly.
Decision tree (challenger)	Readable if/then rules Collections can act on.	Unstable; overfits without depth limits; lower accuracy alone.	Good as a rule-based cross-check.
Neural network / boosted trees (benchmark)	Capture non-linear patterns and interactions.	Black box; needs SHAP; higher governance and monitoring effort; overfits small data.	Only if it clearly beats the simple model and can be explained.

Key trade-off: a small accuracy gain from a complex model rarely justifies losing explainability in consumer lending. The plan is to start simple, benchmark against a complex model, and adopt the complex model only if the gain is material and it passes fairness and explainability checks.

3. Evaluation Strategy

Because only 16% of customers are delinquent, raw accuracy is misleading: a model that predicts "not delinquent" for everyone scores 84% accuracy but catches no at-risk customers. The evaluation therefore uses imbalance-aware metrics, reliability checks and fairness checks.

Metrics and how to interpret them:

Metric	What it tells us	How to interpret
AUC-ROC	How well the model ranks delinquent above non-delinquent customers.	0.5 = random; about 0.7-0.8 acceptable; above 0.8 strong. Primary ranking metric.
Recall	Share of truly delinquent customers caught.	Main business metric: missed at-risk customers mean lost intervention opportunities.
Precision	Share of flagged customers who are truly delinquent.	Shows wasted outreach and unnecessary customer friction; must be weighed against recall.
F1 score	Balance of precision and recall.	Compare models and thresholds on the imbalanced target.
Accuracy	Overall share correct.	Only secondary; always compare to the 84% majority-class baseline.
Confusion matrix, PR-AUC, calibration	Error types, performance on the rare class, whether 20% predicted risk means about 20% actual.	Used to choose the risk-tier thresholds and to check probabilities are trustworthy.

Reliability and validation:

- **Stratified 5-fold cross-validation** plus a held-out test set; report mean and variability of each metric.
- **Baselines:** compare against a majority-class predictor and a simple rule (for example, flag customers with the most missed payments).
- **Stability and drift monitoring after launch:** track the Population Stability Index (PSI), performance by month and recalibration triggers.

Bias and fairness checks:

- **Groups:** age bands, location, employment status and card type (Location and age can act as proxies for protected characteristics, so they are excluded from or tightly controlled in the model).
- **Metrics:** demographic parity difference, disparate impact ratio (flag if below 0.8, the four-fifths rule), equal opportunity (recall gap between groups) and false-positive rate gaps.
- **Mitigation:** remove proxy features, reweight training samples, adjust thresholds per policy review, and use human review for borderline or high-impact cases.
- **Explainability:** publish coefficients and use SHAP for any complex model, so each decision can be justified.

When the model needs improvement:

Improve or pause the model if AUC is below about 0.65-0.70, recall is too low at an acceptable precision, calibration is off, performance differs materially between groups (for example a recall gap above about 5 to 10 points or disparate impact below 0.8), or drift is detected.

Illustrative baseline check on Geldium's data:

As a sanity check, I ran quick 5-fold cross-validated baselines on the Task 1 dataset (median imputation, engineered payment features). Results were close to random: logistic regression AUC 0.48 (recall 0.48, precision 0.17, F1 0.25), a depth-3 decision tree AUC 0.53, and gradient boosting AUC 0.37 with 80% accuracy that comes almost entirely from predicting the majority class (recall 0.03). This matches the weak correlations seen in EDA and suggests that either this dataset has little real predictive signal (it may be simulated) or the Delinquent_Account definition differs from the payment fields. Before deployment, Geldium should confirm the target definition, add richer data (for example longer payment history, balances and transaction behaviour) and re-run the evaluation. These numbers show why AUC and recall, not accuracy, must drive the decision.

Ethical considerations:

- **Fairness:** predictions must not disadvantage groups through direct or proxy variables.
- **Transparency:** customers and regulators can be given clear reasons for a risk flag.

- **Human oversight:** the model prioritises outreach and supports supportive interventions such as reminders and payment plans; it does not automatically penalise customers.
- **Privacy and data minimisation:** use only necessary fields, protect customer data and keep an audit trail.
- **GenAI use:** GenAI drafted the model logic; analysts must validate all outputs against the data and domain knowledge.